batmonitor 0.58-c: one cycle every 5 s | blue = decision (bits + state, make check) | green = matrix / amount | details: charge_flow.pdf, discharge_flow.pdf

inputs (bm_parse: only valid numbers, else stale)
PCC, Sofar Bat1, PV, heat pump W, R290 Hz, BMS SoC / U / CCL / DCL / lowest cell, season, forecast

D1 forecast, B1FIRST, WP_LOW (0.54-c)
-> heat pump row (BM_WP_RULE) LOWSOC / FC / FCBAD / WINTER / TRANS / SUMMER
(z2 = PCC + WP not covered)
-> S3 source DAY / B1 / STK / B1W
make check: 442368 combinations (16 bits x 3 seasons x 8 states)

D5 PCC > 20 kW for 2 min (at once > 23 kW) -> DO4 pulse
pv_relay/DO4 (FoxESS shedding)
first: R290 boost + air conditioning at ≥ 18 kW
grid offline: AC-out feed > 5 kW / unit for 10 s -> at once

S0 caps per stack: chg_cap, dis_cap (charger, BMS CCL / DCL)

heat pump matrix BM_WP_RULE (0.58-c: first match top-down; stacks cover min(share x WP, cap), the rest onto z2) - details wp_matrix.pdf

row	when	rule text	share %	cap W	make check
LOWSOC	lowest stack < WP_SOC_MIN 21 % (back ≥ 23 %)	WP:LOWSOC	0	-	221184
FC	good forecast (stacks above target_soc)	WP:FC	100	-	27648
FCBAD	bad forecast (target_soc 100)	WP:FCBAD	WP_BAT_SHARE_BAD 50	-	10368
WINTER	season winter	WP:Z2	0	-	61056
TRANS	season transition	WP:ALL / T1900	100	WP_BAT_MAX_TRANSITION 1900	61056
SUMMER	season summer	WP:ALL	100	-	61056

orange = share 100 %, no cap: **D6 WPPV** allowed (by day, no D4 cap, PV covers house + WP): z2 += min(50 % WP, 2000 W)

charge path

D4 forecast export cap (May-Aug)
while the forecast expects export > 18 kW:
wait below the cap, charge only above it

D6 WPPV (0.56-c)
WP on PV, row covers all, day, no D4 cap:
chg_ref += min(WP_PV_CHG_SHARE 50 % x WP, WP_PV_GRID_MAX 2 kW)
on surplus > 200 W, held while ≥ 0
make check: 384 combinations

D2 mode per stack, first match:
BMS_MISSING > FORCE > STALE > LADESPERRE > CHARGE / FULL > PROT > CHARGING > DIS
FORCE = SoC < SOC_FORCE or LOW (DCL 0 / cell < 3.05 V; ends at SOC_MIN_RELEASE, 0.55-c)
make check: 16128 combinations per stack (84 bit sets x 12 global x 16 states) -> mode:
BMS_MISSING 8064 · FORCE 5760 · STALE 1152 · LADESPERRE 288 · CHARGE 216 ·
PROT 216 · CHARGING 180 · DIS 180 · FULL 72

D-lead: stack more than SOC_BALANCE_ON ahead

mode matrix BM_MODE_RULE
0 W | FORCE | charge | discharge

charge amount
max(0, own + KP x (z2 + Bat1_eff - target))
chg_ref = z2 (+ D6 share); target + EXPORT_CAP while D4 cap;
≤ sum of chg_cap

FORCE: min(500 W, chg_cap)
any time of day, no TOU window

alloc_levels up to chg_cap
stack by stack in CHARGE_PRIORITY,
lead stack last

discharge branch

BM_SRC_RULE[src]
STK: Bat1 signed, +trickle, idle 0
else: Bat1 charge only, idle 20 W

D3 PROP / IDLE

discharge amount
min(PROP ? deficit : idle, W_MAX, WP_CAP)

alloc_levels up to dis_cap
lead stack first, else
every stack the same share

setpoints per phase -> Multis (+ charge / - discharge)
rule text: mode WP: SRC: split [CAP] [WPPV] [CCL / DCL / WPCAP]
-> state.json -> victron2db.py -> wagodb.pv_victron.bm_l1..3_rule (every minute)

covers all
wp_row
block
no cap
+ share
surplus
SoC
src, z2
CHARGE
FORCE
DIS
0 W rows